Date: 25-10-26

728: Self Dividing Numbers

Approach

1. Initialize result list
2. Traverse the range
3. Copy the number
4. Extract each digit
5. Check divisibility
6. Remove the last digit
7. Add valid numbers
8. Return the result

Time Complexity
O(RD)
R = right - left + 1
D = it is the maximum no. of digits

Auxiliary space

Eg: left = 12 and right = 15
check num = 12
Step 1:- Extract last digit
12 % 10 = 2
check 12 % 2 == 0 → true
Step 2: Remove the last digit
12/10 = 1
Step 3: check the next digit
1 % 1 == 0 → true
Step 4: Finish
temp = 0, so 12 is add to result

12 ✓
13 → 13 % 3 != 0 break
14 → 14 % 4 != 0 break
15 → 15 / 5 == 0 ✓

result [12, 15]

Code:-

```
import java.util.*;
public List<Integer> selfDividingNumbers(int left, int right){
    List<Integer> result = new ArrayList<>();
    for (int num = left; num <= right; num++){
        int temp = num;
        while(temp > 0){
            int digit = temp % 10;
            if (digit == 0 || num % digit != 0){
                break;
            }
            temp /= 10;
        }
        if (temp == 0){
            result.add(num); }
    }
    return result;
```

Problem List

Description | Accepted | Editorial | Solutions | Submissions

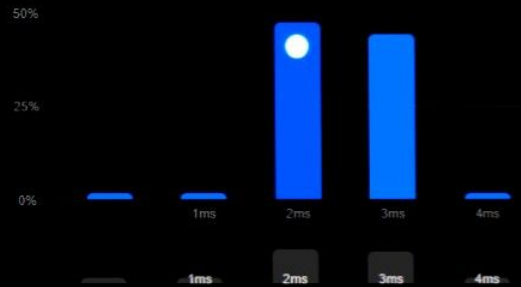
All Submissions

Runtime

2 ms | Beats 98.59%

Memory

42.03 MB | Beats 95.10%



Code Java

Submit 0 Premium

Code

Java Auto

```java
        List<Integer> result = new ArrayList<>();
        for(int num = left; num <= right; num++){
            int temp = num;
            while(temp > 0){
                int digit = temp % 10;

                if(digit == 0 || num % digit != 0)
                    break;
                temp /= 10;
            }
            if(temp == 0){
                result.add(num);
            }
        }
        return result;
```

Saved Ln 13, Col 27

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2